Assignment 5: K-Means / Hierarchical Clustering on Sales Data

Dataset: Sample Sales Data Dataset (Kaggle)

Determine the number of clusters using the elbow method.

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from scipy.cluster.hierarchy import linkage, fcluster, dendrogram
```

Load the dataset

In [2]:
```
df = pd.read_csv("sales_data_sample.csv", encoding="latin1")
df.head()
```

Out[2]:

	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	SALES	O
0	10107	30	95.70	2	2871.00	
1	10121	34	81.35	5	2765.90	
2	10134	41	94.74	2	3884.34	
3	10145	45	83.26	6	3746.70	
4	10159	49	100.00	14	5205.27	

5 rows × 25 columns

In [3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2823 entries, 0 to 2822
Data columns (total 25 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   ORDERNUMBER       2823 non-null   int64  
 1   QUANTITYORDERED   2823 non-null   int64  
 2   PRICEEACH         2823 non-null   float64
 3   ORDERLINENUMBER   2823 non-null   int64  
 4   SALES             2823 non-null   float64
 5   ORDERDATE         2823 non-null   object 
 6   STATUS            2823 non-null   object 
 7   QTR_ID            2823 non-null   int64  
 8   MONTH_ID          2823 non-null   int64  
 9   YEAR_ID           2823 non-null   int64  
 10  PRODUCTLINE       2823 non-null   object 
 11  MSRP              2823 non-null   int64  
 12  PRODUCTCODE       2823 non-null   object 
 13  CUSTOMERNAME      2823 non-null   object 
 14  PHONE             2823 non-null   object 
 15  ADDRESSLINE1      2823 non-null   object 
 16  ADDRESSLINE2      302 non-null    object 
 17  CITY              2823 non-null   object 
 18  STATE             1337 non-null   object 
 19  POSTALCODE        2747 non-null   object 
 20  COUNTRY           2823 non-null   object 
 21  TERRITORY         1749 non-null   object 
 22  CONTACTLASTNAME   2823 non-null   object 
 23  CONTACTFIRSTNAME  2823 non-null   object 
 24  DEALSIZE          2823 non-null   object 
dtypes: float64(2), int64(7), object(16)
memory usage: 551.5+ KB
```

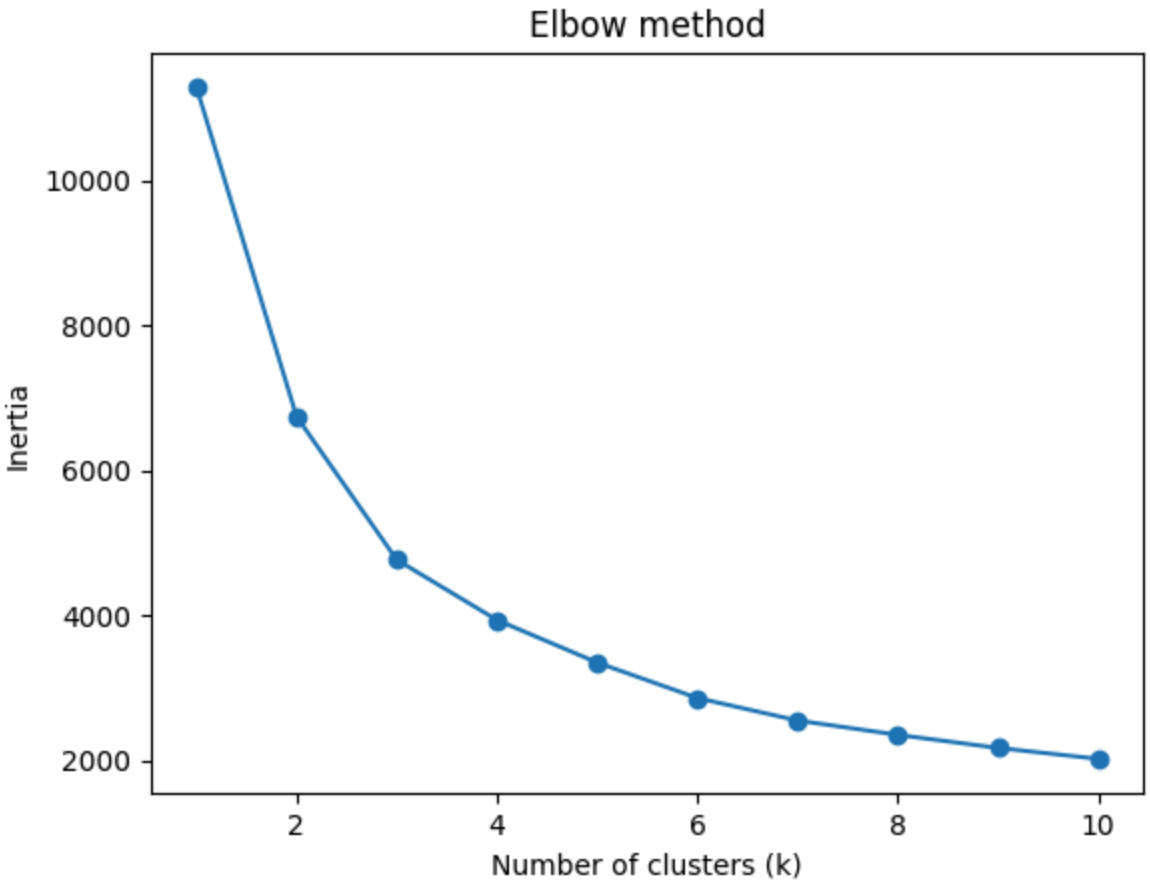
Select numeric features and scale

In [4]:
```
features = ["QUANTITYORDERED", "PRICEEACH", "SALES", "MSRP"]
df_clean = df.dropna(subset=features).copy()
X = StandardScaler().fit_transform(df_clean[features])
```

Elbow method

In [5]:
```
inertias = []
for k in range(1, 11):
    km = KMeans(n_clusters=k, n_init=10, random_state=42).fit(X)
    inertias.append(km.inertia_)

plt.plot(range(1, 11), inertias, "o-")
plt.xlabel("Number of clusters (k)")
plt.ylabel("Inertia")
plt.title("Elbow method")
plt.show()
```



In [6]:
```
diffs = np.diff(inertias)
second_diffs = np.diff(diffs)
best_k = int(np.argmax(second_diffs)) + 2
best_k = max(2, min(best_k, 10))
print(f"Selected number of clusters (elbow heuristic): k = {best_k}")
```

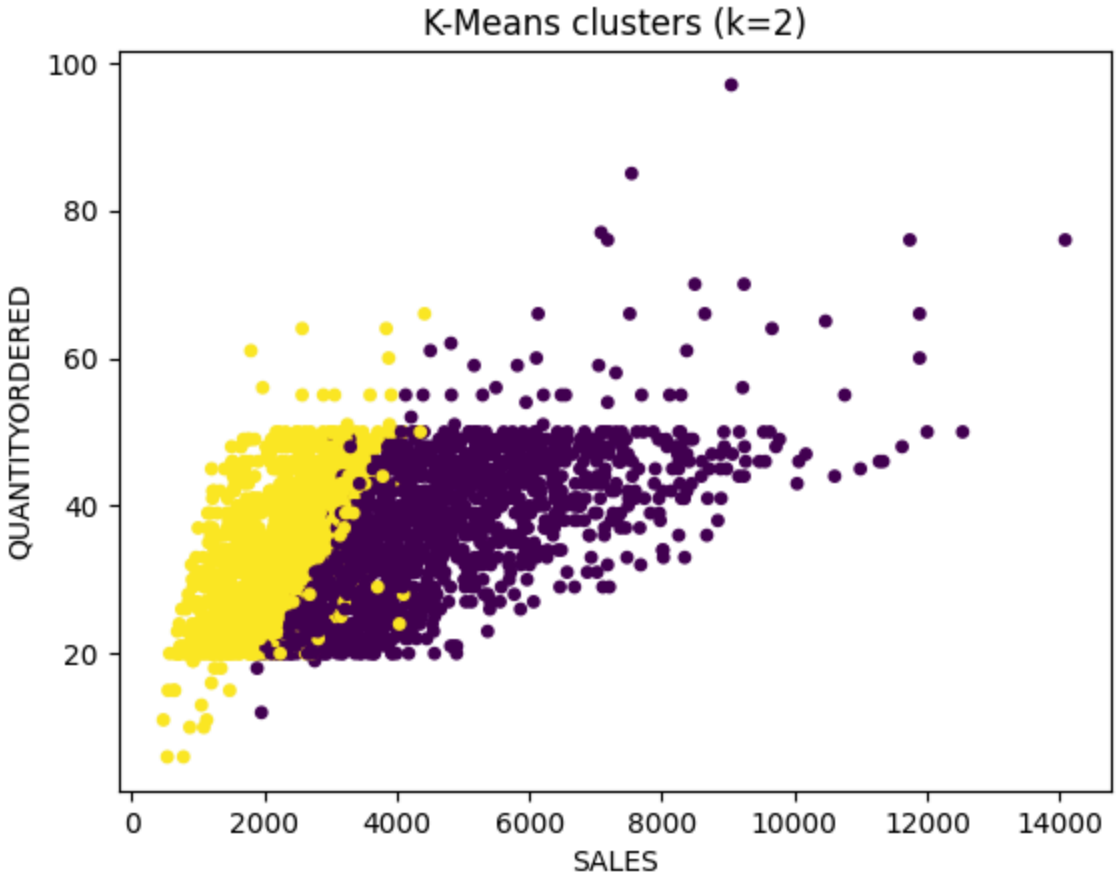
```
Selected number of clusters (elbow heuristic): k = 2
```

Fit K-Means with the chosen k

In [7]:
```
kmeans = KMeans(n_clusters=best_k, n_init=10, random_state=42).fit(X)
df_clean["kmeans_cluster"] = kmeans.labels_
df_clean["kmeans_cluster"].value_counts().sort_index()
```

Out[7]:
```
kmeans_cluster
0    1597
1    1226
Name: count, dtype: int64
```

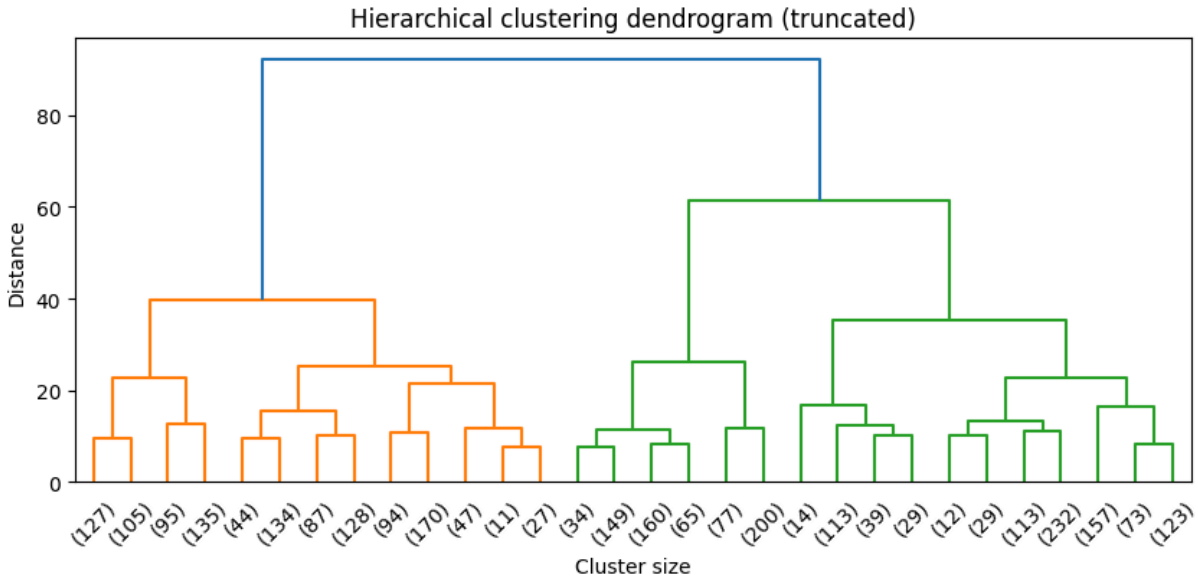
In [8]:
```
plt.scatter(df_clean["SALES"], df_clean["QUANTITYORDERED"],
c=df_clean["kmeans_cluster"], cmap="viridis", s=15)
plt.xlabel("SALES")
plt.ylabel("QUANTITYORDERED")
plt.title(f"K-Means clusters (k={best_k})")
plt.show()
```



Hierarchical clustering comparison

In [9]:
```
linked = linkage(X, method="ward")

plt.figure(figsize=(10, 4))
dendrogram(linked, truncate_mode="lastp", p=30, show_leaf_counts=True)
plt.title("Hierarchical clustering dendrogram (truncated)")
plt.xlabel("Cluster size")
plt.ylabel("Distance")
plt.show()
```



In [10]:
```
hierarchical_labels = fcluster(linked, t=best_k, criterion="maxclust")
df_clean["hierarchical_cluster"] = hierarchical_labels
df_clean["hierarchical_cluster"].value_counts().sort_index()
```

Out[10]:
```
hierarchical_cluster
1    1204
2    1619
Name: count, dtype: int64
```